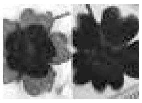
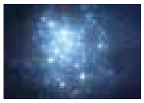


Flowers change colours
Flowers have adapted to rising temperatures altering UV pigments in their petals. **http://dgit.in/oct20-41**

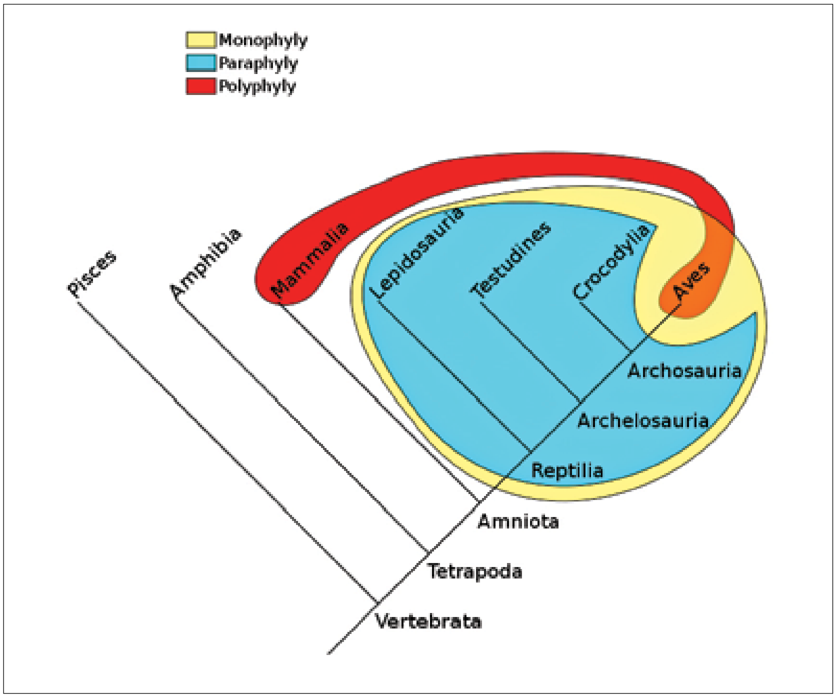


Most luminous galaxy discovered
A galaxy named BOSS-EUVLG1 is the most luminous, almost un-obscured star-forming galaxy. **http://dgit.in/oct20-42**

are by convention, not fish, such as bats and tarsiers. If you take all the creatures with backbones, and leave out all the creatures with four appendages, you get fish.

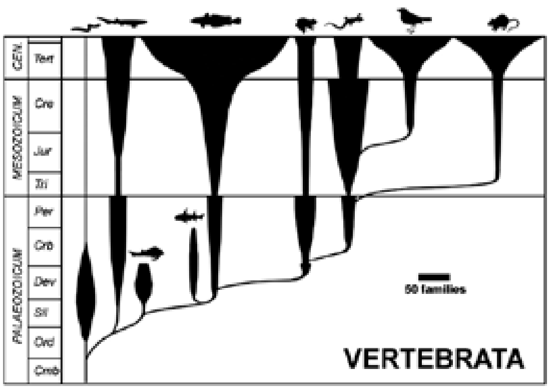
Here, since all the remaining vertebrate classes diverged from bony fishes, all of them should technically also be a fish. So frogs, snakes, humans and horses are all fish. The other option is equally ridiculous. If bony fish are not considered fish, then well, you get a situation where the piranha is technically not a fish. There is no real way around this conundrum, which is why according to scientists, there is no such thing as a fish.

Throughout history, the only real criteria for what seems to be a fish is that it lives in water. There are a number of organisms that bear the name of fishes, but are not really fishes at all. The cuttlefish are molluscs more closely related to squid and octopuses. The starfish are echinoderms, the crayfish are crustaceans, and the jellyfish belong to the phylum Cnidaria. Then there is the shellfish, which includes a wide variety of molluscs, echinoderms and crustaceans. None of these are fish. In the sixteenth century, natural scientists considered seals, whales, amphibians, crocodiles and hippopotamuses as fish. Even today, in the business of fisheries, any aquatic animal that can be harvested is considered a fish, which can include molluscs and crustaceans. This has led to the ridiculous situation where scientists have to use terms such as real fish and fin fish to distinguish fish that are fish from fish that are not really fish, a problem compounded by the fact that there really is no such thing as a fish.



Phylogenetic groups. A monophyletic taxon (yellow) consists of a common ancestor and all its descendants. A paraphyletic taxon (cyan) excludes some descendents from a common ancestor. A polyphyletic taxon (red), such as Haemothermia, which consists of all warm blooded terrestrial animals, does not contain the common ancestor to all its member organisms.

At the heart of the problem are two types of groupings in taxonomy, that are not conducive to scientific study, and as such are discouraged by the scientific community. These types of groupings of taxons are paraphyly and polyphyly. A paraphyletic group consists of the last common ancestor to all the creatures in a group, but arbitrarily excludes a few descendants of the common ancestor. This is the case with fish, where amphibians, mammals, reptiles and birds are excluded. Many species of birds can find it advantageous to lose the ability to fly because of convergent evolution, and so the group of flightless birds is polyphyletic. A monophyletic group is also known as a clade. The problem of excluding certain creatures is common in biology. For example, prokaryotes exclude their descendant group, eukaryotes. The problem of paraphyly is not restricted to fish. For example, there is also no such thing as a worm. Spiders, turtles, wasps and tigers are all worms in the taxonomic rank that includes all the creatures that are considered to be worms, which is a clade known as Nephrozoa. Similarly, reptiles exclude birds, lizards exclude snakes, wasps exclude ants or bees, and moths exclude butterflies. Reptiles and birds make up a clade called Sauropsida, lizards and snakes are Squamates, wasps, ants and bees are Apocrita, and Lepidoptera consists of both moths and butterflies. Crustaceans exclude insects, and monkeys exclude apes. What all of this means is that not only is there no such thing as a fish, there are no such things as reptiles, lizards, wasps, moths, crustaceans and monkeys.



A spindle diagram showing the seven classes of vertebrates. A thicker line means more families.